Programming in .NET
Paper Code: BCAPDCO502T22

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Course outcomes:

At the completion of the course, a student will be able to:

1. Remember the programming skills and be familiar with programming environment.
2. Apply the concept so the students will be able to use ASP.NET controls in web applications.
3. Interpret the to debug and deploy ASP.NET web applications
4. Describe to create database driven ASP.NET web applications and web services
5. To develop, implement, and demonstrate Component Services, Threading, remoting, Windows services.
6. Create and develop Assemblies and Deployment in .NET, Application Development.

Detailed Syllabus

Unit I

The Framework of .Net: Building blocks of .Net Platform (the CLR, CTS and CLS), Features of .Net, deploying the .Net Runtime, Architecture of .Net platform, Introduction to namespaces & type distinction. Types & Object in .Net, the evolution of Web development.

Unit II

Class Libraries in .Net, Introduction to Assemblies & Manifest in .Net, Metadata & attributes. Introduction to C#: Characteristics of C#, Data types: Value types, reference types, default value, constants, variables, scope of variables, boxing and unboxing.

Unit III

Operators and expressions: Arithmetic, relational, logical, bitwise, special operators, evolution of expressions, operator precedence & associativity. Control constructs in C#: Decision making, loops. Classes & methods: Class, methods, constructors, destructors, overloading of operators & functions.

UNIT-IV

Inheritance & polymorphism: visibility control, overriding, abstract class & methods, sealed classes & methods, interfaces.

UNIT-V

Advanced features of C#: Exception handling & error handling, automatic memory management, Input and output (Directories, Files, and streams).

Textbooks:

1. Introduction to C# using .NET By Robert J. Oberg, PHI, 2002,
2. Programming in C# by E.Balaguruswamy, Tata McGraw Hill,

Reference Books:

1. The Complete Guide to C# Programming by V. P. Jain.